CHAOS-ENHANCED PROTOYPICAL NETWORKS FOR FEW-SHOT MEDICAL IMAGE CLASSIFICATION

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

On a broader scale, advancement in medical image analysis through deep learning has attracted considerable attention in recent times, particularly for enhancing diagnostic accuracy in healthcare-related tasks. However, many image-based tasks in healthcare settings are associated with limited labeled data, which affects the performance of conventional deep learning models. Recently, few-shot learning has been recognized as an efficient solution to such problems, which enables a model to learn from a small set of training data. In this paper, a chaos-enhanced few-shot learning-based approach has been proposed for brain tumor classification from MRI images. A combination of a CNN-based model and a prototypical network has been employed to implement a classification-based approach in a low-data regime. Instead of incorporating conventional data augmentation techniques, controlled chaos-based perturbation has been employed in the feature embedding space to enhance the robustness of the model. Various levels of chaos intensity have been investigated to assess their effect on classification performance.The experimental results show that the performance in feature discrimination and classification accuracy can be effectively improved by the moderate chaos perturbation, while excessive chaos results in performance degradation. The effectiveness of the proposed approach in achieving better class separation can also be demonstrated through visualization methods such as t-SNE and confusion matrix analysis.

Keywords: Few-Shot Learning, Brain Tumor Classification, Prototypical Networks, Chaos-Based Feature Perturbation, Medical Image Analysis, Deep Learning, MRI Classification

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ABBREVIATIONS

MRI	Magnetic Resonance Imaging
DL	Deep Learning
PN	Prototypical Network
IoMT	Internet of Medical Things
ReLU	Rectified Linear Unit

NOTATION

N	Number of samples in a dataset
d	Dimension of feature embedding
c	Number of classes
λ	Chaos intensity parameter

Chapter 1

INTRODUCTION

Significant advancements in medical imaging have greatly impacted the diagnosis and monitoring of diseases in modern medicine. Techniques such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT), and X-ray allow doctors to visualize internal body structures. Of all these methods, MRI is commonly used for neurological diagnosis because it is able to clearly visualize internal brain tissue without using harmful radiation. Nevertheless, interpreting medical images is an advanced and time-consuming task that requires skilled radiologists.

To overcome this challenge, the concept of few-shot learning has come into focus. The idea behind few-shot learning is to learn new classes based on only a small set of samples. Unlike other machine learning concepts, few-shot learning focuses on learning generalized feature representation. The idea is to learn feature representation in such a way that it can be generalized for new data. In this regard, few-shot learning is very effective for medical imaging tasks.Among various concepts of few-shot learning, Prototypical Networks have come into focus. The idea behind Prototypical Networks is to map all images onto a feature space where each class has its representative, known as a prototype. The idea is to compare new data with prototypes and predict the class based on the closest prototype. The closest prototype represents the class to which new data belongs. The idea is to achieve classification based on only a small set of data.However, the performance of these models relies significantly on the quality of the learned feature representations. Images in medical diagnostics are prone to noise, different environmental conditions, and structural variations. This might compromise the stability of the class prototypes.

In order to enhance the quality of the learned feature representations, different ideas have been borrowed from nonlinear dynamics and chaos theory. Chaotic systems are known to generate unique patterns in response to small changes in initial conditions. When chaos-based perturbations are incorporated into a machine learning model, they can provide variations in the learned feature space.In this work, chaos theory is used to incorporate feature perturbation in prototypical networks for few-shot medical image classification. This is done to enhance the robustness of the embedding space and improve classification performance when only limited training samples are available.The remaining sections of this report cover background concepts, related work, modelling, experimental evaluation, and conclusions.

1.1 BACKGROUND

In the modern healthcare sector, medical imaging is of significant importance as it enables the doctor to visualize the structures of the human body without the need for any invasive procedures. Medical imaging techniques such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT), ultrasound imaging, and X-ray imaging assist in the early diagnosis of various diseases. Among the various imaging modalities, MRI is employed in the analysis of soft tissues. In the analysis of the brain's structure, MRI is employed as detailed visualization of the brain is necessary. As the number of patients being treated in the healthcare sector is increasing in hospitals, the requirement for image analysis is also rising.

In the traditional approach, the images obtained through various imaging modalities are analyzed manually by the radiologists and other expert medical professionals. Although the analysis of images by expert professionals is highly accurate, the process is often associated with certain drawbacks. In the busy healthcare sector, the radiologists often have to analyze a number of images in a limited period of time.Earlier approaches to computer vision involved different feature extraction techniques like edge detection, texture analysis, and statistical analysis. Although these techniques are helpful in finding some patterns in medical images, they are not sufficient to find all the patterns present in medical images.Recently, deep learning has revolutionized image analysis techniques. Deep neural networks, particularly convolutional neural networks, are able to automatically learn different features directly from images. These networks are able to identify different patterns that are not easy to identify using conventional techniques.However, deep learning models usually require large annotated datasets. Collecting large datasets in medical domains is challenging due to privacy issues, inaccessibility of patient information, and requirement of expert knowledge for labeling. Furthermore, certain medical conditions may be rare, leading to a small number of instances in a dataset. These issues may make it challenging for traditional deep learning models to learn reliable models.

To overcome this limitation of traditional deep learning models, researchers have investigated learning techniques that can be used even when a small amount of data is available. Few-shot learning has attracted significant attention in this context. Few-shot learning aims to learn general feature representations that can be used for adapting to new classes even when a small number of instances is available. Collecting large datasets for medical imaging is challenging.Among the various methods of few-shot learning, prototypical networks have demonstrated significant promise for classification problems. These models learn a space where all instances of a class are mapped close to a representative instance, known as a prototype. For classification, a new image is classified based on how similar it is to these prototypes. This enables the network to learn new classes even when it is provided with a small number of instances.Overall, the renewed focus on few-shot learning demonstrates that there is a requirement for intelligent systems that can assist medical practitioners when training data is limited.

1.2 Challenges in Medical Image Classification

Medical image analysis faces several unique constraints that differentiate it from standard computer vision tasks:

- **Subtle Morphological Changes:** The pathological markers can be microscopic in nature. It is important to differentiate between healthy tissue and early-stage disease, which can only be achieved by identifying subtle changes in texture or intensity, which can be difficult to detect by most algorithms.

- **Expert-Dependent Data Scarcity:** Unlike other data sets, medical image data sets require expensive and time-consuming annotation by experts in radiology. This creates an issue of scarcity in medical image data sets, resulting in much smaller training sets compared to natural image data sets.

- **High Inter-domain Variability:** Variability in hardware (scanner brands), protocols, and physiology of patients also creates noise. A model trained on data from one hospital may not generalize well on data from another hospital due to "domain shift."

- **Inherent Class Imbalance:** Clinical data is naturally imbalanced, with healthy data far outnumbering rare pathological conditions. This results in biased models that lean towards the majority class, possibly missing life-saving diagnoses.

- **Acquisition Artifacts:** Physical factors such as patient movement during an MRI or sensor noise cause distortions. the Differentiating between a genuine lesion and a motion artifact is one of the most important technical challenges.

- **The Need for Clinical Explainability:** In addition, accuracy alone is insufficient in a clinical context. In order for it to be viable, it also needs to be "interpretable" in such a way that doctors can follow the rationale behind the prediction.

- **Emerging Solutions:** The Current research is focused on few-shot learning and robust feature a extraction to address the data scarcity and reliability issues in the diverse clinical the environments.

1.3 Problem Statement

This explosion of medical image data has led to a huge need for accurate diagnostic tools, but there's a significant problem. Deep learning, as a field, has shown incredible promise for image classification, but it requires a huge number of images to actually work well. In the medical field, this is a huge problem. The issue of getting accurate labels requires a significant number of doctors taking time away from practice to manually go through images and accurately label them. As a result, there's a huge need for a solution that can effectively work with very few images.

Of all the methods that have been proposed, prototypical networks have shown particular promise as a solution. The idea behind them is very simple and intuitive. Instead of relying on traditional neural networks, a prototypical network works by effectively "mapping" a class to a "prototype" and then classifying images based on how close they are to that prototype. However, there's a significant problem. The issue of sensor noise, as well as natural differences between patients, can throw off the accuracy of this prototype, leading to a significant downturn in performance. The purpose of this project is to create a solution that works well even in a scenario where there are few images available. To this end, there's a significant need to incorporate chaos-inspired perturbation into a prototypical network. The idea behind this is that, by effectively "stress-testing" the network, a more accurate solution can be developed.

Chapter 2

LITERATURE SURVEY

2.1 Limitations of Conventional Medical Image Classification

Well, medical image classification has certainly made some leaps forward in the recent past with the advent of machine learning and deep learning. The majority of them follow the conventional supervised learning paradigm in which the models are simply fed massive datasets and then learn [1]. CNNs, or Convolutional Neural Networks, are pretty much the de facto standard when it comes to feature extraction for things such as MRI scans, pathology slides, or even X-rays [2]. However, when they are actually deployed in the real world in a hospital setting, it's a whole different ball game because of the rather significant constraints [3].

The most important constraint that conventional methods face is that they simply require massive datasets. However, in the medical domain, this is just a complete and utter pain. For instance, there are privacy concerns and the fact that one needs an expert radiologist to actually sit there and label all the data [4]. Consequently, the datasets that are actually available are rather small and imbalanced. This makes it rather difficult for the conventional deep learning approach to actually generalize well. In the event that they are actually deployed with such small datasets, they simply overfit and fail to perform well in the real world[5].

Another thing is that medical images tend to be complex and messy. The visual differences that allow one disease to be distinguished from another tend to be extremely subtle[6]. Moreover, they also tend to vary depending on the subject, brand of machine used, or even settings. Such subtle differences often make it extremely difficult for standard classification techniques to pick up on them, especially if there is only limited training data for that particular class[7]. This is even more evident if you are dealing with rare diseases, where you might only have a handful of representative images to work with. To make matters worse, these techniques tend to be "data hogs" that demand huge amounts of computational power in order to be trained from scratch[8]. This makes it extremely difficult for them to be used in clinical environments where hardware is limited. Most standard techniques also tend not to be flexible in adapting to new classes or conditions. However, in a dynamic medical environment, it is extremely important that models be able to update their knowledge without having to go back into training[9]. All of these issues point towards the fact that we need an entirely new approach that is capable of working with fewer labels but still being reliable. This is exactly what recent research is focusing on—meta-learning and few-shot learning techniques that focus on helping models learn how to transfer knowledge from one task to another[10].

2.1.1 Traditional Machine Learning Approaches

Before deep learning took over, medical image classification was primarily done using what we can term "traditional" machine learning. Now, before deep learning took over, medical image classification was primarily done using what we can term "traditional" machine learning[11]. Now, before deep learning took over, medical image classification was primarily done using what we can term "traditional" machine learning. Now, before deep learning took over, medical image classification was primarily done using what we can term "traditional" machine learning. Now, before deep learning took over, medical image classification was primarily done using what we can term "traditional" machine learning.

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The biggest headache, though, was that there was too much reliance on ”handcrafted” features. It takes a ton of expertise and lots of experimentation to get a feature that actually works. Moreover, there is often too much emphasis on the subtle features or high-level features in an image, especially if the difference between a healthy tissue and a diseased tissue is extremely small[14]. Another major problem with the traditional techniques is that they were not scalable. As medical images got bigger and more complex, these traditional features were not able to cope with all the rich features in the images[15]. Furthermore, there was too much emphasis on selecting features that were appropriate for traditional classifiers. It was extremely time-consuming, and it did not generalize well if you wanted to apply it on a different set of images or even on a different type of scan[16]. All of these were essentially what led everyone to adopt data-driven models that we have today, which could also learn features on their own. However, it is also important to understand where we have come from, since we essentially used traditional techniques that laid the foundation for what we have today[17].

2.2 Few-Shot Learning Approaches

Few Shot Learning, or FSL for short, has really taken off as a big deal in machine learning lately, especially for domains where you simply can't get your hands on a huge dataset. The idea behind this technique is to help your model learn how to "catch on" to a new classification task with a mere handful of examples. It's kind of like how humans work, where you can typically identify a new class of objects after seeing them only once or twice. This technique is a total lifesaver for medical imaging. Getting thousands of images annotated is simply impossible, especially considering all the red tape, laws, and the fact that you need a highly compensated expert to sit down and annotate every single image[18].

While traditional supervised learning models are data hungry and require thousands of images, few shot learning focuses more on the idea of knowledge transfer from one task to another. In the training phase, the model goes through a series of "episodes," where it's shown a "support set" and then a "query set." The support set gives the model a few images to learn from, and then the query set tests whether or not the model "gets it." Repeating this process time and time again for different tasks, the model eventually learns a way of representing features that works for brand new tasks.

Recently, this type of learning has witnessed a ton of interest in medical image analysis. Researchers are playing around with prototype-based and meta-learning frameworks to detect disease categories from limited samples. These types of learning are super useful if you're dealing with rare diseases, where maybe only a few patients in the entire world have been scanned. By concentrating on these "transferable" features instead of relying on huge samples, few-shot learning is offering a much more reasonable way of developing medical classification systems. Of course, it is not perfect, and there are still plenty of challenges [19]. For instance, there are still plenty of challenges in making the feature extraction process even more effective, with the accuracy remaining high with only two or three images.

2.3 Prototypical Networks for Few-Shot Learning

I think that the most popular approach that people take when it comes to few-shot learning with the use of metrics is the one that is called the prototypical networks. The whole purpose of this approach is that it represents each class with this thing called the prototype. The prototype is simply the average or the mean feature embedding of just a small number of the support examples that were provided in that class [20]The purpose of this approach is that it is just the right one when it comes to few-shot learning because it provides one with the ability to classify things in the simplest way possible. The approach does not attempt to learn complex decision boundaries like the standard approach. It simply looks at the distance that is represented in the features.

The way it does this is that the embedding network is trained on tons of small tasks and learns how to produce features that cluster naturally. Recently, there has been a lot of research on how to make the original prototypical approach even better. This is particularly true when it comes to medical image analysis. For instance, researchers are working on better prototype representations, multi-channel feature extraction, and better regularization methods to make the classification more stable even when the data is scarce. Research by Quan et al. and Ouahab and Ahmed have proven that these prototype-based architectures can actually perform quite well with pathology and medical datasets with just a handful of samples.

The simplicity of the approach and the strength of the approach make the prototypical approach the go-to approach when it comes to building the foundation of most modern few-shot methods. It's hard to argue with the simplicity and accuracy of the approach when working in a clinical setting.

2.4 Limitations of Existing Few-Shot Learning Methods

While few-shot learning sounds fantastic on paper, there is still much work to be done in dealing with all the issues that plague these techniques in applying them to complex medical imaging tasks. The first major problem is that it is simply not easy to "discriminative" features with only a handful of training images at your disposal. When dealing with extremely small amounts of data, there is often not enough detail in the images for your model to pick up on what makes one particular disease unique from another [21]. Another issue that often plagues metric-based techniques is that the class prototypes that your model uses do not always remain entirely stable. Since your model is usually only able to calculate these prototypes using only a handful of images from your set of support images, it is not always entirely representative of all the ways that a particular disease could manifest in patients.

Then there is the optimization and meta-learning side of things, which brings a lot of technical complexity to the table. These types of methods require very specialized training strategies and thousands of episodes to actually yield good results [22]. It is also quite expensive in terms of computational power and is very sensitive to hyperparameters. If the hyperparameters are off by even a little bit, the whole system will not converge properly.

However, there is also the problem of the fact that most of the few-shot learning models do not actually think about feature diversity and robustness. This is particularly problematic in the medical field because factors such as image quality, different machine brands, and even movement of the patients can totally change the image. Most models do not account for these types of factors yet, and all of these problems illustrate the importance of finding ways to improve the robustness of the models, such as through feature refinement, optimization, or even those chaos-inspired methods to improve the prototypes even when data is scarce [23].

Chapter 3

Problem Statement And Methodology

3.1 Overview and Objectives

So, this chapter is basically just describing the modeling setup that we're using for our few-shot medical image classification. The whole point is to create something that actually works for classifying these different medical images, even if there's not really any data available to work with. Most deep learning models require massive amounts of data in order to work properly, but in medicine, that's actually a huge problem. Not only is it difficult due to privacy issues, but it also requires an actual expert to go through and classify every single pixel in an image. The whole point of this system is that it uses few-shot learning and specifically prototypical networks in order to classify things.

So, basically, the logic behind this model is that it tries to find "good" features in the images that will make it easier to classify them. Rather than trying to classify everything and create hard lines between different classes, it tries to put everything into this small space called an embedding space. This is helpful because it means that it will be able to classify new kinds of diseases and images even if it's only been exposed to a small number of them.

For the actual architecture, we are using a Convolutional Neural Network (CNN) as our main feature extractor. This CNN works with our raw images and reduces them down to their simplest form. So, it's picking up on textures, edges, or those weird spots you see in medical imaging. These are then used to generate something called "prototypes." These are really just averages of things in a certain category during an episode of training.

The actual training is done in these episodes, which is slightly different from regular training. You have your support set and your query set. Your support set is what gives it the examples it needs to generate these prototypes. Your query set is what's used to test it and see if it can actually match things correctly based off of distance. Thousands of these episodes really help it get good at similarity judgments across many different types of medical imaging. It's a pretty efficient way of working, and it really helps make this framework much more reliable in real-world applications, especially considering how scarce data is. It may not be 100% perfect for every edge case, but it's a good start.

3.2 Dataset Description

At the end of the day, the quality of the data that the machine learning system receives is what ultimately determines the quality of the system. In the case of medical imaging, the system needs to be able to pick up on small visual cues that might indicate that there is a serious problem. In the process of creating this project, we used a specific medical imaging dataset to build our few-shot classification system. The whole idea is to provide the system with enough diversity that it can learn how to distinguish one thing from another, even if it hasn't been provided with very many examples of that thing. The datasets that we're working with come from real-world sources in hospitals and come in the form of MRI scans, CT scans, and even X-rays. The data itself is pretty complex because it is depicting the inner structures of the body that can vary depending on the settings of the machine and even the position of the patient during the time of the scan. So, in order to build a quality system, we need to be able to take all of this diversity into consideration so that the system learns the biological features and not just general noise.

The dataset itself is organized in such a way that there are different diagnostic groups that have been made. In each image that we're working with, there is a tag that says what condition the image is depicting. Essentially, this is the "truth" that we're working with in this process.

We've organized the files in such a way that if all the images in the same category are clustered together, it makes it way easier to load the data and set up these training "episodes." Essentially, the system will just repeat and repeat and repeat, pulling small batches of images to make the support and query sets over and over again. It's like simulating a real-life scenario in which the doctor might only have a couple of previous cases to compare the new scan against. However, before all that even occurs, we have to do some preprocessing. The medical images can be all over the place in terms of brightness and/or size and/or angle because of the different equipment that was used. So, we've taken a few steps to normalize all of that so that the network can just focus on the pathological features instead of getting sidetracked by the weirdness of the image. This dataset is the basis upon which the model can learn features that actually generalize well.

3.2.1 Medical Image Dataset Characteristics

To be honest, working with this medical dataset is a whole different ballgame compared to standard computer vision problems such as sorting images of cats or cars. It's much trickier because the signs of a disease can be super subtle, such as a tiny change in texture or a tissue shape that's slightly off, which you would normally overlook. I think that's why the way we structured the dataset is so important because, in a way, it's the only way the model can learn to pick up on those specific signs instead of guessing.

The images are arranged in such a way that they're grouped by the condition, with each condition having its own label. Each image is assigned a tag that the process can use to connect the images with a specific condition in the real world. It's much easier to store them in such a way because it's easier to pull images for the few-shot learning process, which is important because we don't have a huge amount of data to work with. I think what's most striking about this dataset is how varied the images are. Each person is uniquely different, and no two scanning devices are ever perfectly calibrated in the same way. This means there are images that are too bright, some that are blurry, or some that are flipped, in addition to the random noises from the scanning devices themselves. A good system needs to be able to filter all of that "junk" out and only focus on the specific features of the images.

The biggest headache, though, is the unavailability of labeled data. It takes forever to get a busy specialist to sit down and label all these images. Moreover, in the case of rare diseases, you can't get heaps of samples anyway. This is why we're pushing for few-shot learning in the first place—to get the model to learn from just a few images and still get it to work on new cases. The images also contain various stages of illness, from early onset to later stages. While this helps the model learn about the illness better, it also means that the network can get "distracted" by weird things in the backgrounds instead of the important areas of the images. Overall, though, this is a very realistic testing ground for the few-shot model, though we still have to get through the preprocessing steps to make things more uniform before we can start the training process.

3.3 Few-Shot Learning Framework

The basic idea behind this system is to find a way to do the classification of the medical images well, even if we are stuck with a really small number of samples. For most real-world clinical scenarios, getting huge datasets is a total nightmare, since the process of annotating the data is super slow and requires experts. As a result, the "usual" data-hungry deep learning models just don't work well in this case. To get around this problem, the few-shot learning framework is implemented, which is designed to learn the feature representations using just a few samples. As opposed to the usual models, which would force the mapping between the images and the labels using the huge datasets, the few-shot learning framework learns the so-called "embedding space" where the images are naturally grouped together. Here, the classification is done using the distance between the feature representations.

The few-shot learning framework is implemented using the task-based learning strategy, which is pretty common in few-shot learning research. As part of the training process, the system is shown many small classification tasks that are representative of the ones that would be encountered at the testing stage. For each of the tasks, the system is shown just a few examples of each class, which the system has to use to learn the patterns that would enable it to classify the new, unseen samples belonging to those classes. This is great since the system is forced to learn the actual patterns and understand the similarity between the classes, rather than relying on the huge datasets. One of the biggest advantages of this system is its ability to generalize well between the classes. As a result, the system is able to learn the features well enough to be able to spot new diseases, even if the system has seen just a few examples of those classes before.

The other significant component of the framework is the prototype-based learning strategy. In essence, what the model does is to learn a representative feature vector, called a prototype, for each of the classes using the support set. It essentially reduces the entire classification process to a measure of distance from the prototypes. It is a significant reduction of the entire process, but the model's capacity for generalization is still strong. We have also introduced a component that introduces chaos in the learning process. It essentially introduces variability in the learning process, which enables the model to learn robust features without having to use conventional data augmentation techniques. By adding chaos to the features, the model can handle unseen images much better. It is a much more efficient way of dealing with the issue of limited medical data, which combines representation learning with prototype classification to arrive at robust results.

3.3.1 Support Set and Query Set Formation

In the few-shot learning paradigm, the formation of support and query sets is an important aspect in the training process. In most traditional machine learning approaches, the entire dataset is employed in the training process. However, in few-shot learning, the data is divided into small tasks to mimic real-world situations. In few-shot learning, the images are divided into two groups: support set and query set. This enables the model to learn to classify images based on the few examples given. The support set contains a few examples for the selected class in a given training episode. These examples are considered to be representatives from which the model learns to identify the characteristics of the class. In prototypical learning, the images in the support set are passed through the feature extraction network to obtain the embeddings in the latent feature space. These embeddings are then employed to obtain the prototypes for the classes. The prototypes are considered to be representatives from which the characteristics of the class are learned. The prototypes are considered to be reference points for the support set images belonging to a given class.

On the other hand, the query set consists of images that need to be classified by the model based on the information it learned from the support set. The query samples also go through the same feature extraction network to obtain their feature embeddings. The model now compares the query embeddings with the class prototypes learned from the support set. The class that is closest to the query embedding in the feature space is chosen as the predicted class for the image. This allows the model to perform classification by similarity comparison rather than the traditional decision boundary approach.

In the actual process, the construction of the support and query sets is an iterative process during the training process to generate different classification tasks for the model. This allows the model to learn from different classes and different samples from the data set during the training process. This process allows the model to gain a better understanding of the relationships between images from the same class and those from different classes.The episodic nature of the support and query set construction also plays an important role in enhancing the ability of the system to generalize. This is due to the fact that the system is presented with a number of small tasks with limited samples. As a result, it focuses on the significant feature representation as opposed to memorizing the samples. This ability is significant in medical imaging applications, as there are situations where there are limited examples for certain medical conditions.On the whole, the construction of support and query sets enables the few-shot learning model to mimic real-world classification problems.

This enables the system to classify medical images effectively, as it is capable of identifying the pattern and making accurate predictions even when there are limited examples for a given medical image category.

3.4 Backbone Feature Extractor

Another important part of the suggested few-shot learning scheme is the backbone feature extractor. The primary function of the backbone feature extractor is to map the medical images into a numerical space that can be utilized for classification. There are complex patterns present in medical images that need to be understood and classified for the sake of medical diagnosis. These complex patterns present in the medical images are difficult to understand and process with the help of a simple algorithm. Therefore, a deep neural network is utilized for the automatic extraction of features from the medical image.The backbone network is a feature extraction module that takes an image and maps it into a compact feature space. Instead of working with the pixel values of the image, the system maps the image into a compact space that contains the most important features of the image. This provides a more meaningful representation of the image and allows the few-shot learning scheme to classify the image based on the similarity of the features present in the image.

In this research, a deep convolutional neural network is utilized as a backbone feature extractor. Convolutional neural networks are popular in image processing tasks, and they are powerful in learning features from images. Features in early convolutional neural network layers often learn basic image features such as edges and textures, while later layers learn more abstract and high-level features from images. This feature learning mechanism of convolutional neural networks is useful in identifying fine differences in visual features among classes of medical images.Once an image is fed into a backbone convolutional neural network, the feature vector is a representation of an image in an embedding space where similar images have similar feature representations. This is a basis for the image classification using a prototype-based method in few-shot learning. By comparing feature representations of query images with those of a support set of images, a classification of an image is possible.

Another significant benefit of employing a deep feature extractor is that it is able to generalize over various classes. This means that if a new class is encountered by the model with limited samples, it is still able to learn a feature representation that is able to identify certain similarities between images. This is a very useful property in medical image analysis, where there is limited labeled data available.In addition, it is evident that the backbone feature extractor is a crucial component of the proposed framework, where it is utilized to convert raw medical images into useful feature embeddings that enable the few-shot learning model to classify images based on similarity comparison, rather than relying on a large dataset of images. The following subsections discuss the architecture of the feature extractor and the feature embeddings generated from the input images.

3.4.1 ResNet Architecture

For effective feature extraction in medical images, the proposed framework utilizes a deep convolutional neural network based on the Residual Network. Deep neural networks have been widely known to learn complex patterns in images. However, as the depth of the neural network increases, training the neural network becomes challenging. Some of the challenges include the occurrence of vanishing gradients and degradation of performance. To overcome this problem, the ResNet architecture has been introduced. In this type of neural network, residual connections are introduced to allow information to flow easily in the neural network.

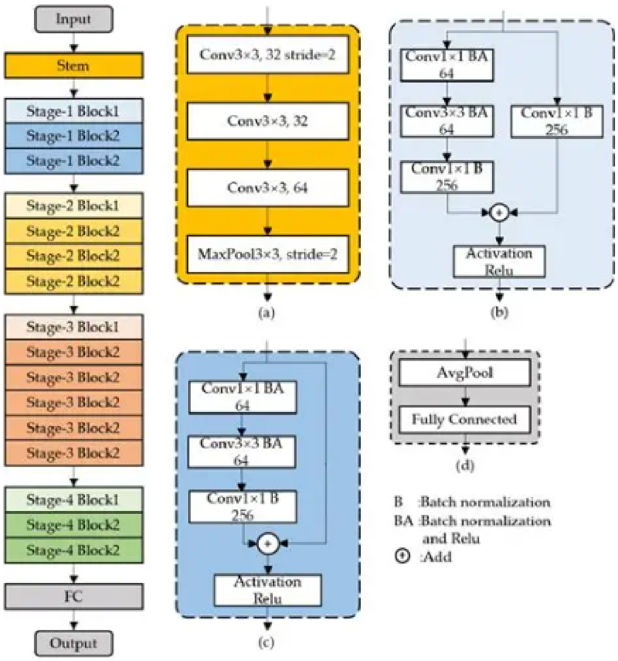

Figure 3.1: ResNet Architecture

The idea behind the ResNet neural network is to use residual learning blocks. In a deep neural network, each layer in the neural network tries to learn the direct mapping between the input and the output. However, in a residual neural network, each layer in the neural network learns the residual mapping between the input and the output. This means that each layer in the neural network learns the difference between the input and the output. This has been achieved by introducing shortcut connections in the neural network. In this type of neural network, shortcut connections are introduced to allow information to flow easily in the neural network.Within this proposed system, the ResNet model acts as a feature extractor that transforms medical images into feature representations. In this case, the ResNet model processes each image in the dataset using a number of convolutional neural networks. As the image progresses through this process, it learns to recognize hierarchical features. Initially, the image may learn to recognize edges and intensity gradient features. However, as it progresses to the later stages of the neural network, it becomes proficient in recognizing complex features, including structural arrangements and abnormalities that may be associated with medical conditions.

Another advantage of this proposed system, in relation to the ResNet model, stems from its capability to generate feature embeddings. In this case, feature embeddings represent the essential features of the medical images. By using this feature embedding, the system eliminates redundant information. In the few-shot learning framework, this feature embedding plays a critical role in creating prototypes. In this case, the feature embeddings of similar images have similar characteristics. Therefore, the few-shot learning framework can use this information to make classifications.The ResNet architecture also fits in seamlessly with the episodic training approach employed in the few-shot learning framework. In the episodic training approach, during each training episode, both support and query images are passed through the same backbone network to obtain feature embeddings. This consistency in feature extraction for both support and query images enables reliable distance-based comparison between the two.Overall, the employment of the ResNet-based backbone network provides a solid foundation for the proposed classification system. The ability to extract deep hierarchical features and the stability provided by residual connections make it possible for the proposed classification system to obtain accurate embeddings for few-shot classification of medical images. The following subsection discusses the process by which these embeddings are obtained and employed in the proposed learning framework.

3.4.2 Feature Embedding Generation

Once the medical images have been processed by the backbone network, the next phase is to obtain feature embeddings, which are a numerical representation of the images. Feature embeddings are a vector representation of images that captures their fundamental characteristics while avoiding the high dimensionality of image pixels. Feature embeddings are considered the fundamental elements of few-shot learning, where comparisons and classification of images are performed.In few-shot learning, an image is fed into a backbone network, such as a ResNet, where a series of convolution, activation, and pooling operations are performed on the image in various stages. Each of these stages progressively abstracts the image by identifying various patterns, such as edges, textures, shapes, and relationships, in the image. As the image moves further into the network, the features become more abstract and more relevant to classification. The final output of the backbone network is a feature vector, which is a representation of the fundamental features of an image, such as edges, shapes, and textures, that are visible in the image.

This feature vector, which is usually called an embedding, represents the image in a lower-dimensional feature space. In this feature space, images that belong to the same category are closer to each other in the embedding space, while images that belong to different classes are further apart in the feature space. This is a very critical aspect of the few-shot learning task, as the model compares the similarity of the images rather than using a large dataset to determine the class of a new image.In the proposed framework, the model calculates the embeddings for both the support set and the query set images in each episode of the training process. The embeddings of the support set images are used to calculate representative prototypes for each class. These prototypes are essentially reference points for each class in the embedding space. Once the query image is processed, its embedding is compared with all the prototypes of the classes, and the class is determined based on the closest prototype.Another significant aspect of incorporating generation in this piece of work is the incorporation of a chaos-driven mechanism. Instead of using conventional data augmentation mechanisms, a certain level of perturbations is added in the feature space during the training phase of the network. This, in turn, slightly alters the embeddings in a systematic manner, enabling the network to learn more stable feature representations.

Through this embedding generation mechanism, raw medical images are converted to feature representations that are informative and contain a certain level of visual structure. This is where the raw medical images are classified based on their visual features using a classification mechanism known as a prototype-based classification mechanism, which is incorporated in the proposed few-shot learning system.The proposed few-shot learning system is based on a prototypical network, and in this section, the architecture of this network is discussed, explaining how the embeddings generated in the previous section are utilized to classify raw medical images based on their visual features.

In order to maintain numerical stability and ensure consistent feature scaling, the embeddings produced by the backbone feature extractor are normalized prior to prototype computation. Normalization helps control variations in embedding magnitude and allows similarity comparisons to rely more on the geometric structure of the feature space.

Let $f_\theta(x)$ denote the embedding produced by the feature extraction network for an input sample x. The normalized representation z is obtained as

$$z = \frac{f_\theta(x)}{|f_\theta(x)|_2 + \epsilon} \tag{3.1}$$

Where $|f_\theta(x)|_2$ represents the L2 norm of the embedding vector and ϵ is a small positive constant introduced to avoid division instability when the norm approaches zero. This normalization step projects the embedding into a bounded feature space, which improves stability during prototype formation and distance-based classification.

Chapter 4

System Design

4.1 Prototypical Network Architecture

The prototypical network acts as the main component in the classification process for the proposed few-shot learning framework. The prototypical network framework is different from traditional classification models that are learned from complex classification boundaries. The prototypical network framework focuses on learning a feature space in which all images from the same class are grouped. This framework is considered to be highly effective for medical image analysis tasks, as there are limited samples available for training.In the prototypical network framework, a class is represented by a prototype, which is the average feature representation of the support samples in the respective class. In the prototypical network framework, during the training process, the support images are first processed through the backbone feature extractor to obtain the feature embeddings. The feature embeddings are then combined to obtain a prototype for the respective class. The obtained prototypes are the average feature representations for the respective classes.

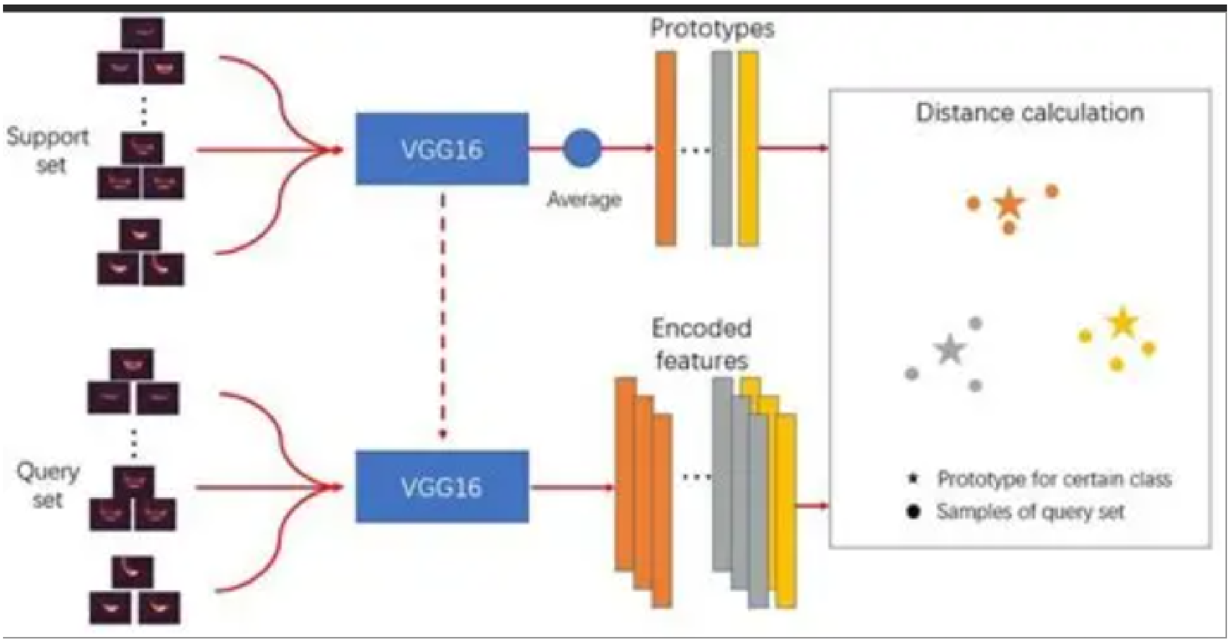

Figure 4.1: Prototypical Network Architecture

Once the prototypes are computed, the classification is performed as a similarity comparison. For the query images, the same network is used to compute their feature embeddings. Then, the distance is measured between the query's embedding and the prototypes of all classes. The class whose prototype is closest to the query's embedding is selected as the predicted class for the given query image. The distance-based decision mechanism enables the model to classify images well even when there are few images for each class.The prototypical network's architecture is highly advantageous as it facilitates the simplification of the learning mechanism while retaining its strong generalization power. As the model is able to learn to organize images in the embedding space, it can easily adapt to new classes that may not have been well-represented during training. This is highly advantageous in medical applications, as new classes may appear with limited training data.

By representing classes as prototypes, the system can classify images based on similarity. This helps in ensuring a robust classification system even when operating in a limited data regime. The detailed process of computing prototypes, along with the classification mechanism, is discussed in the following subsections.

4.1.1 Prototype Computation

The computation of the prototype is an important aspect of the functioning of the prototypical network, which is used in the proposed framework for few-shot learning. In this framework, each class is represented as a prototype, which is used as a reference point in the feature embedding space. Instead of using a large number of labeled images, a small number of support images are used to form a representative prototype for each class used in a training episode.The first step in the proposed framework is to pass all the images of the support set through a backbone feature extractor. As discussed in the previous sections, the backbone network transforms each image into a feature embedding, which represents the important features of each image. These feature embeddings are numerical representations of each image in a lower-dimensional feature space where the similarity between images can be measured more effectively.

Once the embeddings are obtained for the support images, the prototype for the class is obtained by averaging the embeddings belonging to the class. Averaging is performed to obtain one single feature vector, which is the central tendency of all the support images belonging to the class. Therefore, the obtained vector is the prototype for the class, and it is obtained as the average of all the support images belonging to the class. By doing this, the class is represented using only a few support images, and at the same time, there is meaningful information in the features.The prototype can be seen as the center of the cluster formed by the embeddings of the images belonging to the class. When there are several support images, averaging their embeddings helps to reduce the effect of the noise and variations present in the images.During the process of episodic training, the computation of prototypes occurs repeatedly for all tasks. Due to the varying classes and examples selected in every episode, the prototypes are dynamically updated throughout the training process. This helps the model learn an embedding space where images from the same category naturally form clusters around their corresponding prototypes.

In addition, in the proposed framework, the chaos-driven mechanism interacts with the process of computing prototypes. This causes a slight variation in the embedding space, which affects the generated embedding. This variation helps the model learn more robust classes, which are not affected even when a small variation exists.This process enables the computation of a simple yet effective summary of the class information in the embedding space by the process of prototype computation. This enables the classification process to be performed by the model even when the number of training samples is low by summarizing the class information into a single vector that is derived from the support samples for each class. The next subsection discusses the process of classification by the prototypes for the query images.

4.1.2 Distance Metric Computation

After the class prototypes are obtained from the support set, the next step in the prototypical network framework is to ascertain the process by which query images are classified. This is done by comparing the distance between the embeddings obtained for the query images and the class prototypes. The main concept behind this classification process is that images from the same class are likely to be closer to their respective class prototypes in the feature space.At this stage, the process begins by passing the query image through the same backbone network as the support set. This ensures that both support and query images are embedded in the same space. The output from the network is a feature embedding that encapsulates the characteristics of the query image. Once the embedding is obtained, the distance between the obtained embedding and all the class prototypes in the set is measured.A suitable distance metric is used to measure the similarity between the query embedding and each of the class prototypes. In most prototypical network architectures, the Euclidean distance is frequently used as it is simple and effective in measuring the similarity between two data points in the embedding space. The result of the distance operation represents the extent to which the query image shares features with a given class.

Once the distances between the query embedding and all the class prototypes have been obtained, the model then uses them to generate a probability distribution over classes. In most cases, a softmax operation is performed on the negative distances to ensure that the class with the minimum distance is given the highest probability. The class with the minimum distance is then used to determine the query image class.This classification strategy based on distance is particularly suitable for few-shot learning tasks. This is because, compared to more complex classification strategies, this model can easily accommodate new classes even when only a few instances are available. As long as a reasonable feature representation is obtained from the feature extractor, classification of new instances can be performed based on similarity comparison with available prototypes.In addition, within the context of the discussed framework, another effect of the chaos-driven mechanism is seen in terms of influencing the embedding space. This effect causes a more discriminative representation of the images. Consequently, even when some variation exists in the images, a reasonable distance can always be obtained when comparing a query image to a given prototype. This variation ensures a more discriminative representation of images. Overall, the computation of the distance metric represents the final component of classification within the prototypical network. By considering similarity between query images and prototypes, a reasonable classification can be obtained to identify the best class for a given image.

4.2 Chaos-Based Feature Perturbation

One of the interesting features of the proposed framework is the inclusion of a chaos theory-inspired perturbation mechanism for the features. Usually, in traditional deep learning models, data augmentation techniques such as rotation, flipping, scaling, and intensity variation are employed to improve the variability of the training samples. Although such techniques improve the generality of the models, they need to be designed and implemented by the users and may not effectively handle the variability of the medical image data. Instead of employing the traditional data augmentation techniques, the proposed approach uses a chaos theory-inspired perturbation technique to improve the variability of the features.According to the chaos theory, a deterministic system can be defined as a system that is capable of producing complex and sensitive variations. The chaos theory states that a small change in the initial condition of the system can result in a drastic variation in the system's behavior. The proposed approach employs the chaos theory to introduce variations in the features learned by the backbone network without actually introducing variations in the input images.

Once the feature embeddings are generated by the backbone network, during the training phase, the chaos-based mechanism slightly modifies the feature embeddings by mathematically controlled transformations. This does not significantly change the feature space, and this change is just enough to make the model learn more stable and discriminative feature representations. This makes the model less sensitive to slight variations in the data and provides better generalization capabilities to the model.The addition of chaos-based perturbations provides a similar effect to implicit augmentation. Instead of generating different versions of an image, this variation is added to the feature embedding space where classification is performed. This reduces the need to design augmentation pipelines while still providing the model with a variety of training scenarios.

Another advantage of the proposed mechanism is that it can be utilized with the episodic training strategy that is a part of the few-shot learning approach. Since the number of samples is fixed in the training process, the variability introduced at the feature level prevents the network from overfitting the training set.Overall, the chaos-based feature perturbation mechanism increases the robustness of the proposed system by introducing variability during the training process. The direct manipulation of the feature space makes the proposed mechanism compatible with the architecture of the prototypical network and enables the effective utilization of the few-shot learning approach without the need for traditional image-level variability.

To introduce controlled variability during episodic training, a deterministic chaotic signal is generated using the logistic map. The logistic map is widely used in nonlinear dynamical systems to model chaotic behavior and is suitable for generating bounded perturbations.

The logistic chaos sequence is defined as

$$x_{t+1} = r, x_t(1 - x_t) \tag{4.1}$$

where x_t represents the value of the chaotic sequence at iteration t, and r is the control parameter that determines the dynamical behavior of the system. When the value of r lies in the chaotic regime (typically $3.57 < r \leq 4$), the generated sequence exhibits sensitive dependence on initial conditions and produces non-repeating patterns.

Let x_t denote the chaotic value obtained from the logistic map at iteration t. The scaled chaotic perturbation δ_t is computed as

$$\delta_t = \lambda, x_t \tag{4.2}$$

Where λ represents the noise scaling coefficient that regulates the strength of the perturbation. By adjusting this parameter, the influence of chaotic noise on the embedding space can be carefully controlled.

The z_s denotes the normalized embedding corresponding to sample s, and Noise represents the scaled chaotic perturbation.

$$\hat{z}_s = z_s + \text{Noise} \tag{4.3}$$

Where $\hat{z}_s$ represents the final embedding used for prototype formation and distance-based classification. By injecting the perturbation into the embedding space, the model is encouraged to adapt to slight variations in the feature distribution, which improves robustness during episodic learning.

4.2.1 Logistic Map Chaos Theory

As the series progresses, even small variations in the initial condition lead to significant variations in the output, a phenomenon that is characteristic of chaotic systems.In this regard, in the context of the proposed model, a logistic map is utilized to introduce certain perturbations in the feature embedding space. Instead of directly altering the original medical images, a chaotic sequence is generated by employing a logistic map and is utilized to make certain modifications to the feature vectors, which are originally generated by a backbone network. Such modifications to the feature embeddings enable the model to learn to handle certain variations in the data, without directly employing traditional image augmentation strategies.

The utilization of a logistic map in this regard is also advantageous in certain ways, such as providing a simple yet effective mathematical framework to introduce certain variations in the data, and owing to its deterministic nature, the proposed model is able to introduce certain variations in the data while still ensuring a certain level of reproducibility in the data, which is often required to learn stable feature representations by a deep neural network.Another key characteristic of the logistic map is its ability to achieve a certain range of values. This characteristic makes it possible to control the range of perturbations introduced to the feature embeddings. By choosing appropriate parameters in the logistic equation, the range of perturbations can be controlled to ensure it remains small enough not to interfere with the learning process but remains significant enough to introduce variability.The integration of logistic map chaos in the feature learning process enables the system to mimic some of the possible variations that may exist in real-world medical imaging situations.

4.2.2 Chaos-Enhanced Feature Embeddings

At this stage, the aim is to introduce a certain level of variation in the feature space, which is learned by the model, without altering the original medical images provided to the system. This ensures that, during the learning phase, the model is able to learn a stable feature representation that is less likely to change in response to variations in the original images provided to it.In the proposed framework, a feature embedding is generated for a medical image by applying a backbone feature extraction mechanism to it. This feature embedding is a numerical representation of the visual features of the image provided to the system. Instead of directly using this feature embedding to compute prototypes and classify medical images, a certain level of variation is introduced to it, generated by applying a chaotic sequence generated by the logistic map to the feature vector in a controlled manner, so that the information embedded in it is preserved.

After generating the perturbed embeddings, a representative prototype is computed for each class using the samples from the support set. The prototype serves as the central representation of a class in the embedding space and is later used for distance-based classification.

Let S_c denote the support set corresponding to class c, containing K labeled samples (x_i, y_i). The prototype p_c for class c is calculated as the mean of the perturbed embeddings belonging to that class:

$$p_c = \frac{1}{K} \sum_{(x_i,y_i)\in S_c} \hat{z}_i \tag{4.4}$$

where $\hat{z}_i$ represents the perturbed embedding of the i^{th} sample.

The chaotic values, in this case, act as a modulation factor, slightly changing the position of the embeddings in the feature space. This ensures that the model is able to learn feature representations that are not overly dependent on the actual numerical values of the embeddings. This is a great mechanism, as it allows the model to learn to focus on the general structure and relationships between feature vectors, rather than memorizing certain patterns from a limited set of samples.In a certain sense, this mechanism is an implicit feature augmentation mechanism, where instead of generating a variety of altered versions of the original image, variations are directly applied in the embedding space where classification decisions are made, simplifying the entire training process while still exposing the model to a variety of representation patterns during episodic training.The other advantage associated with chaos-enhanced embeddings is that they contribute to the enhancement of robustness in the system. The medical images may come with small variations due to various factors such as differences in patient anatomy and various noises that might be introduced during the imaging process. The system becomes more efficient in handling these variations when generating embeddings for the input images during the testing process.The proposed system has enhanced the robustness of the system in representing the input data using the integration of chaos-based perturbation in the process of generating embeddings for the input data. The embeddings retain their discriminative power but become more robust in handling small variations in the input data.

4.2.3 Similarity Computation

Once the class prototypes are obtained, the similarity between a query embedding and each class prototype is calculated. This similarity measure determines how closely the query sample aligns with the learned class representations in the embedding space.

Let z_q denote the embedding of the query sample and p_c represent the prototype corresponding to class c. The similarity between the query embedding and the prototype is computed as

$$\mathrm{sim}(z_q, p_c) = \tau \frac{z_q \cdot p_c}{|z_q|_2 |p_c|_2} \tag{4.5}$$

4.2.4 Loss Function

During training, the model parameters are optimized by minimizing a classification loss computed over the query samples in each episode. The loss encourages the query embedding to be more similar to the prototype of its correct class while reducing similarity with other class prototypes.

Let Q denote the set of query samples, where each query sample is represented as (x_q, y_q). The training loss is defined as

$$L = -\sum_{(x_q, y_q) \in Q} \log \left(\frac{e^{\mathrm{sim}(z_q, p_{y_q})}}{\sum_{k=1}^{N} e^{\mathrm{sim}(z_q, p_k)}} \right) \tag{4.6}$$

Where $\mathrm{sim}(z_q, p_k)$ denotes the similarity between the query embedding z_q and the prototype of class k, p_{y_q} represents the prototype corresponding to the true class label of the query sample, and N is the total number of classes in the episode. This objective encourages the model to assign higher similarity scores to the correct class prototype during training.

4.3 Model Training

Once the architecture of the proposed framework is defined and the mechanism of chaos-based feature perturbation is integrated, the framework is trained to learn discriminative feature representations from the medical image dataset. To this end, the framework is trained in a manner that ensures its capability to classify images accurately even in scenarios where a limited number of samples are available from each class of images. To this end, during the training phase, the proposed framework is trained by incorporating an episodic learning strategy in conjunction with chaos-based feature perturbations.To this end, during the training phase, the proposed framework is trained by iteratively sampling the available dataset to construct classification tasks in an episodic manner. To this end, a limited number of classes are sampled from the available classes, and their images are divided into support and query sets, where the support set is utilized to compute class prototypes, and the query set is utilized to test the proposed framework's ability to classify images based on their similarity to the computed prototypes. Both support and query images are fed into the feature extractor to obtain feature embeddings, which are utilized to compute class prototypes.

One important aspect of the training process in this work is the application of chaos-based feature perturbation at various levels of intensity. In most traditional methods, a fixed level of feature perturbation is considered during the training process. In contrast, in the proposed approach, the training process is carried out by applying chaos-based feature perturbations at various levels, ranging from 0% to 40%. The levels of feature perturbations are increased in steps or intervals, such as 0%, 1%, 2%, 3%, and so on, up to 40%.When the level of chaos is set to 0%, it means that there is no feature perturbation in the system. In this case, the feature embeddings learned by the backbone network are directly considered for the calculation of the prototypes and classification. This level of chaos is considered to check the performance of the system without the application of chaos-based feature perturbations. When the percentage of chaos increases, feature perturbations are applied to the feature embeddings. These feature perturbations are derived from the logistic map and cause slight changes in the feature vectors while maintaining the structure.

The training of the model across various chaos levels is useful in examining the impact of perturbations in features on the model's generalization capability. Training across various chaos levels, from lower to higher, provides diversity in the embedding space. By training and testing across the entire range, from 0% to 40%, it is possible to determine the best chaos level for obtaining the most accurate classification results.At every training iteration, the model calculates the classification loss based on the distance between query embeddings and class prototypes. Then, the loss is utilized to update the network parameters using backpropagation. After several training episodes, the model is able to learn to properly organize the feature space, where images from the same class are grouped together at their respective prototypes, and images from different classes are well separated.Through the use of episodic training in combination with chaos-based perturbations, a robust training process is developed which promotes the creation of robust feature embeddings. This training process enables a robust classification performance even when limited data is available along with small variations in image characteristics.

4.4 Computational Environment

The training and development of the suggested few-shot learning model were performed in a computational environment that is appropriate for the experimentation and training of deep learning models and the effective processing of data. This is because the suggested framework includes the training of convolutional neural networks and the performance of the tasks involved in the process of generating the feature embeddings and the episodic learning process.The implementation of the suggested system was performed by using the Python programming language, which is appropriate for the implementation of machine learning models due to the existence of a wide variety of libraries and tools that can be integrated with the language for the effective implementation of machine learning models.For the construction and training of the neural network model, the PyTorch framework for deep learning was employed. This framework enables the construction of dynamic graphs and efficient computations involving tensors, making it an efficient choice for the construction of few-shot learning prototypes. The framework also enables researchers to create customized training processes and integrate various episodic learning strategies and specialized mechanisms, such as chaotic feature perturbations.

Other Python libraries were also employed for the implementation process. Libraries such as NumPy were employed for carrying out computations and for performing operations involving matrices. The Matplotlib library was also employed for the visualization of experimental results and trends. In addition, libraries such as PIL and Torchvision were employed for carrying out image processing and for managing datasets.The training of the model and experiments were performed using a system that is capable of providing the necessary computational resources to support deep learning computations. A typical setup would involve a multi-core processor, system memory, and optionally, a graphics processing unit (GPU) to speed up computations, as it is capable of executing parallel computations. The usage of a GPU speeds up the training significantly when training convolutional networks and when there are many episodic training tasks.

Chapter 5

RESULTS AND DISCUSSIONS

The experimental results demonstrate that the introduction of chaos-based feature perturbation has a significant impact on the classification performance of the proposed model. As the chaos level is set to 0%, the proposed model learns feature embeddings without perturbation; hence, moderate class separation is achieved in the feature space.

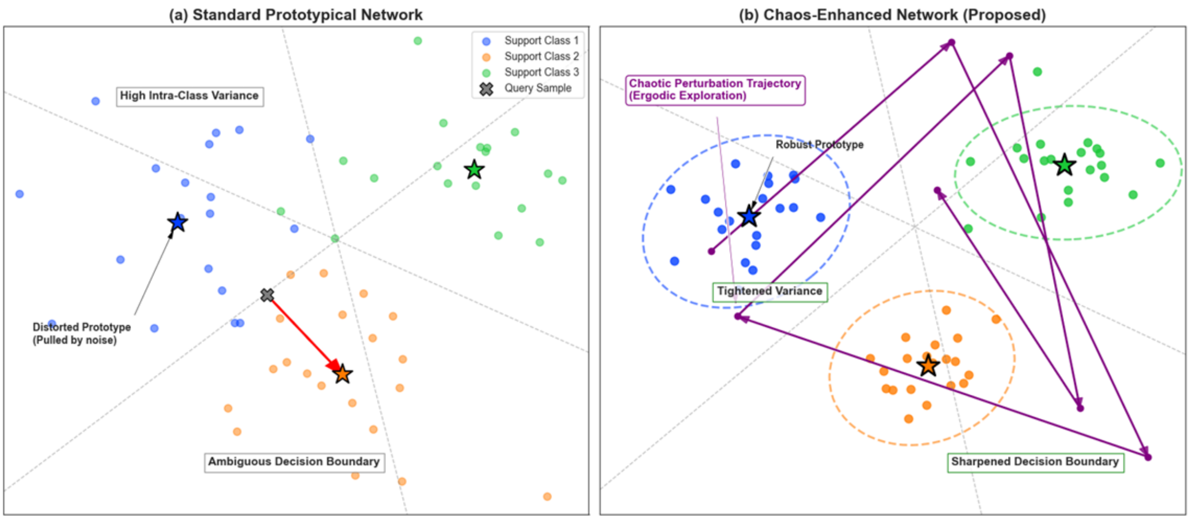

Figure 5.1: Difference Standard prototypical networks and Chaos enhanced netwroks(Proposed).

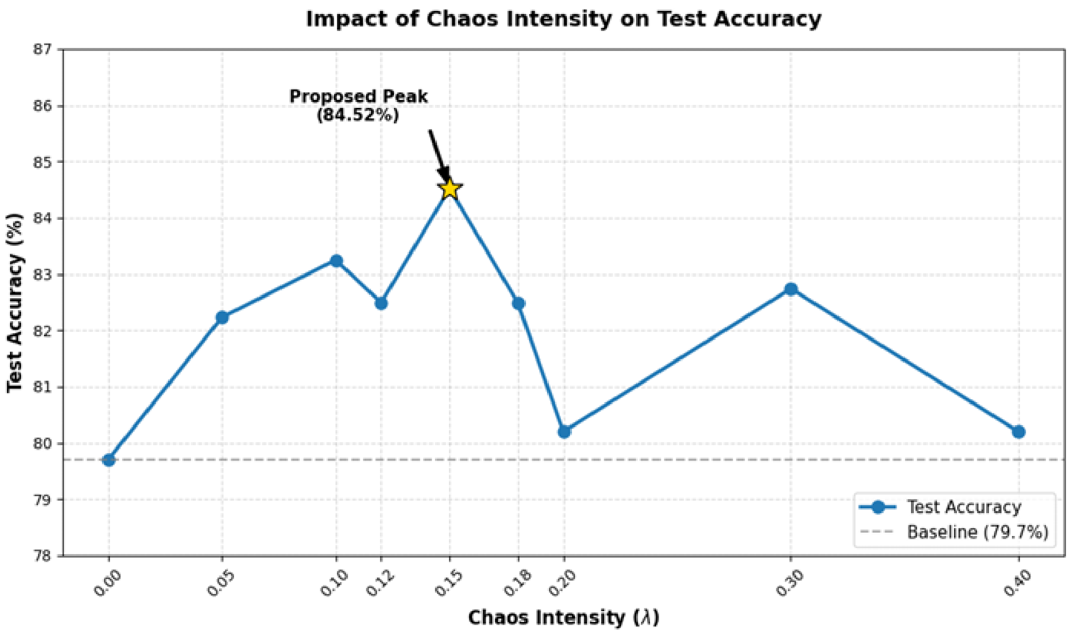

Figure 5.2: Impact of chaos intensity on test accuracy

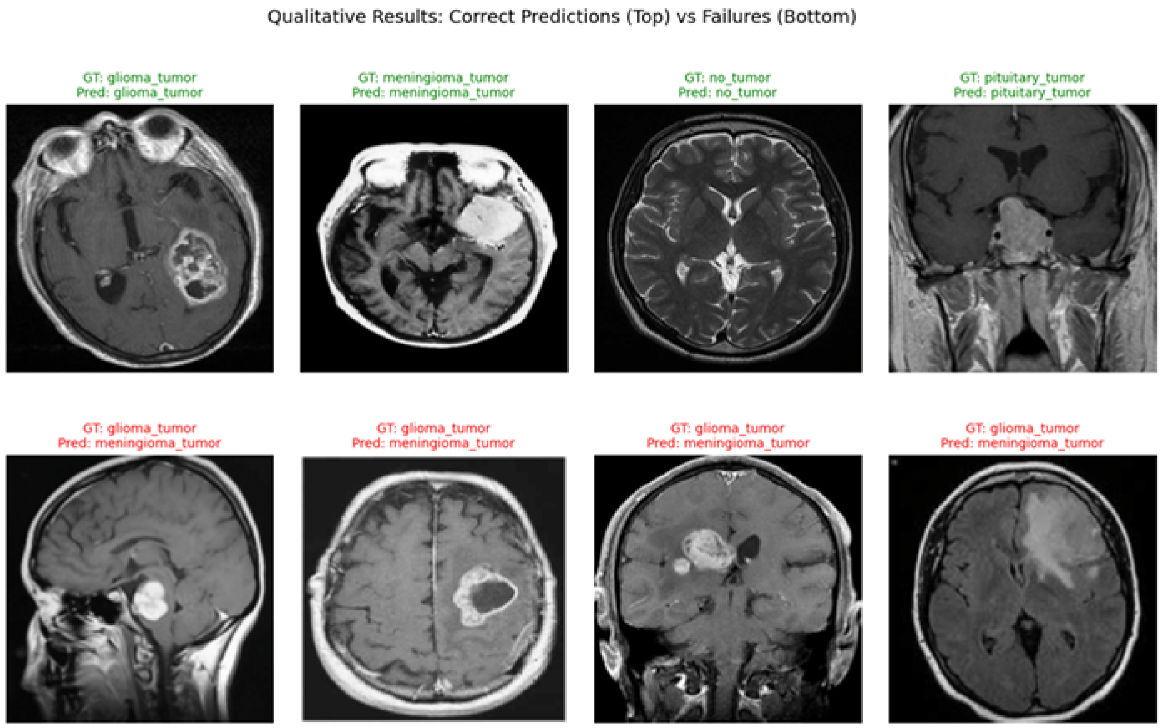

Figure 5.3: The model catches pituitary tumors by spotting sella turcica changes.

To evaluate the influence of chaotic perturbations on the embedding space, experiments were conducted by varying the chaos intensity parameter from 0 to 40. The purpose of this experiment was to analyze the impact of various levels of chaotic perturbations on the stability and separability of the learned feature representations. In the complete set of experiments, three experiments were chosen to be analyzed: chaos level 0, which represents no perturbation; chaos level 15, which represents the optimal level of perturbation; and chaos level 40, which represents excessive levels of perturbation.

5.0.1 Performance at Chaos Level 0

The level of chaos is 0 when there is no chaotic perturbation added to the embedding. The behavior of the model is similar to that of a normal prototype-based learning system without further regularization in this case.

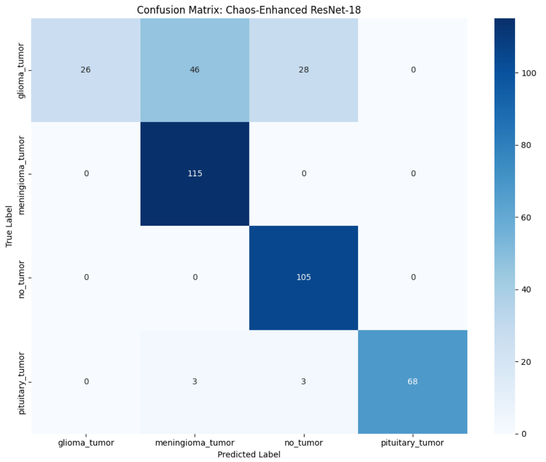

Figure 5.4: Confusion Matrix with Chaos 0

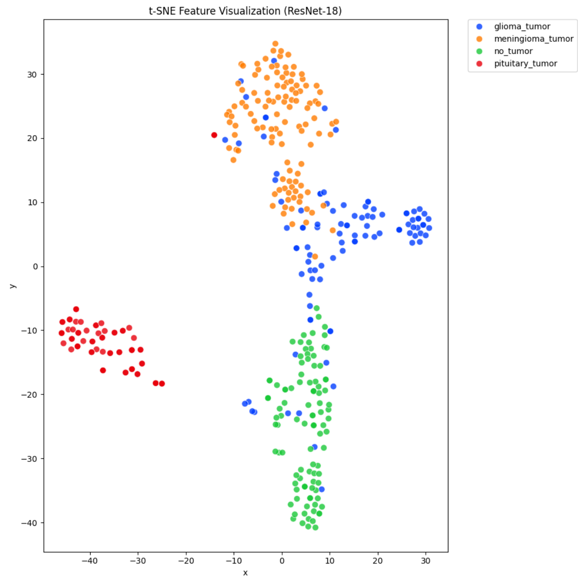

Figure 5.5: t-SNE Visualization with Chaos 0

The confusion matrix in Fig. 5.4 indicates that although several classes are well classified, there are considerable misclassifications in visually similar classes. This suggests that the model is having trouble clearly distinguishing the class boundaries in the embedding space. The t-SNE plot in Fig. 5.5 reinforces this observation. The clusters for different classes seem to be somewhat close to each other. There is even some overlapping in the clusters in some areas. This indicates that the embedding representation is not discriminative enough without the introduction of chaotic perturbations.

5.0.2 Performance at Chaos Level 15

The highest classification accuracy was obtained by chaos level 15. It can be noted that an increase in chaos level up to 15 provided moderate variability in the embedding space, which in turn helped the model learn robust feature representations.

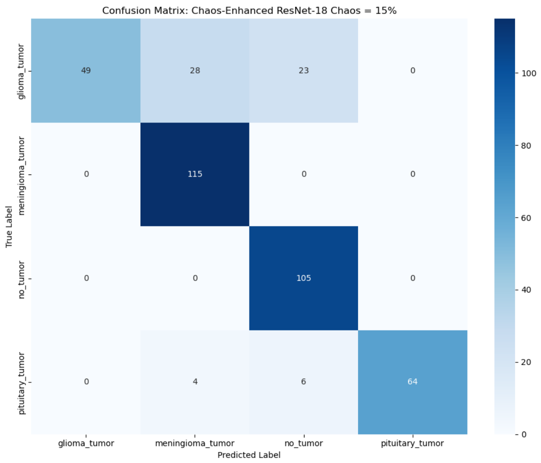

Figure 5.6: Confusion Matrix with Chaos 15

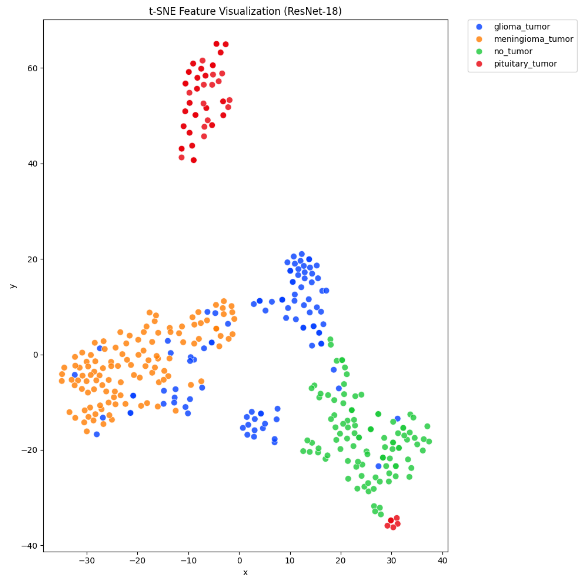

Figure 5.7: t-SNE Visualization with Chaos 15

As depicted in Fig. 5.6, the confusion matrix reveals that there is a significant improvement in classification accuracy compared to the baseline case. The diagonal dominance of most classes is very strong, indicating that the model is capable of correctly classifying samples of each class with higher accuracy.The t-SNE plot in Fig. 5.7 shows that the samples are well separated with minimal overlap between them. The embeddings of the same class are closely packed together, and each class is separate from the others. This is the reason for the higher accuracy achieved during classification at this level of chaos, which is moderate and effective in stabilizing the prototype formation process.

5.0.3 Performance at Chaos Level 40

When the chaos parameter is increased beyond its optimal value, it is observed that the perturbations introduced in the embedding space are too high. This may affect the performance of the prototype-based classification.

From the confusion matrix in Fig. 5.8, it is observed that the number of misclassifications is increased as the chaos level is increased up to 40. Although some classes are still separable, it is observed that many off-diagonal elements appear in the matrix. This indicates that there is confusion between those classes. The t-SNE plot in Fig. 5.9 also indicates that the clusters are again becoming disorganized and overlapping.

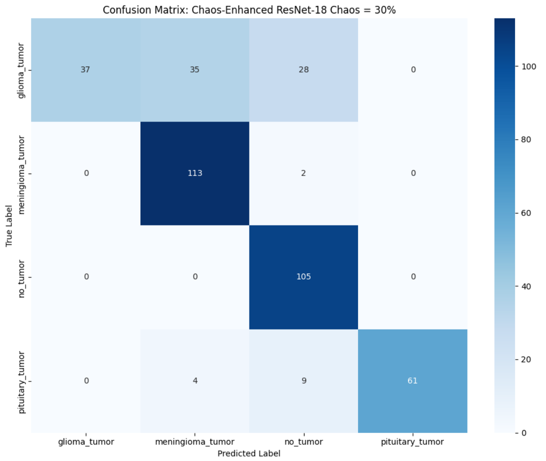

Figure 5.8: Confusion Matrix with Chaos 40

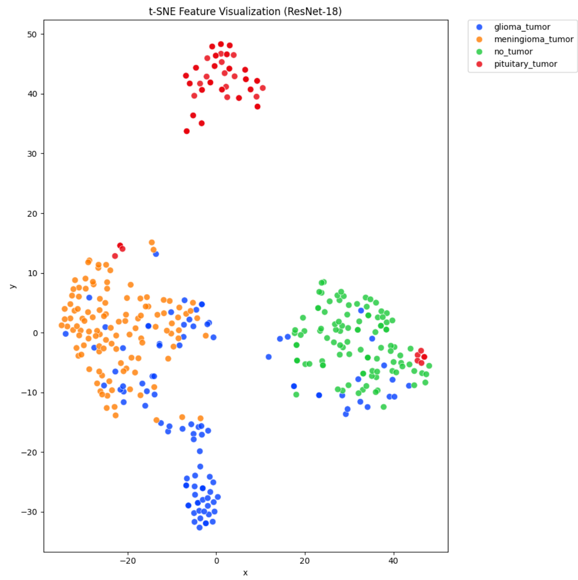

Figure 5.9: t-SNE Visualization with Chaos 40

5.0.4 Discussion

The experimental results demonstrate that there is a strong correlation between the level of chaotic perturbation and the quality of the learned feature representations. It can be noticed that, as the chaos level increases from 0 to 15, the quality of the learned embedding space improves accordingly. This can be explained by the regularization property of the introduced chaotic perturbations, which forces the model to learn more generalized feature representations instead of overfitting to particular patterns.

On the other hand, an increase in the level of perturbations beyond a particular optimal value leads to a decrease in the quality of the learned embedding space. For instance, as the chaos level increases to 40, the perturbations destabilize the learned embedding space, leading to a decrease in the quality of the learned embedding space. Consequently, the quality of the classification drops. From the experimental results, it can be concluded that a moderate level of chaotic perturbations is essential in stabilizing the embedding space to improve the quality of the classification using the proposed framework. Among all the configurations, chaos level 15 produced the best results, with the highest classification accuracy.

Table 5.1: Performance Comparison at Different Chaos Intensity Levels

Chaos Intensity (λ)	**Accuracy (%)**	**Macro Precision**	**Macro Recall**	**Macro F1-Score**
$\lambda = 0.00$ (Baseline)	79.70	0.8683	0.7947	0.7665
$\lambda = 0.05$	82.23	0.8745	0.8145	0.8058
$\lambda = 0.10$	83.25	0.8847	0.8271	0.8190
$\lambda = 0.12$	82.49	0.8805	0.8196	0.8088
$\lambda = 0.15$ (Proposed)	**84.52**	**0.8915**	**0.8387**	**0.8354**
$\lambda = 0.18$	82.49	0.8819	0.8249	0.8047
$\lambda = 0.20$	80.20	0.8608	0.7951	0.7823
$\lambda = 0.30$	82.74	0.8693	0.8222	0.8164
$\lambda = 0.40$	80.20	0.8681	0.7942	0.7834

Table 5.2: Class-wise Performance Metrics

Class Label	**Precision**	**Recall**	**F1-Score**	**Support Samples**
Glioma Tumor	1.0000	0.4900	0.6577	100
Meningioma Tumor	0.7823	1.0000	0.8779	115
No Tumor	0.7836	1.0000	0.8787	105
Pituitary Tumor	1.0000	0.8649	0.9275	74

Table 5.1 and Table 5.2 present the performance analysis of the proposed chaos-enhanced few-shot learning framework for brain tumor classification. Table 4.1 shows the overall per-

formance of the model at different levels of chaos intensity. The baseline model without chaos achieves an accuracy of 79.70 percent. As the level of chaos increases gradually, the classification performance also improves, indicating that controlled perturbation in the feature representation helps the model learn more discriminative patterns. The best performance is obtained when the chaos intensity is set to 0.15, where the model achieves an accuracy of 84.52 percent along with higher macro precision, recall, and F1-score values. This trend suggests that introducing a moderate level of chaos enhances the model's ability to generalize across different classes. However, when the chaos intensity is increased beyond this point, the performance begins to decline slightly, which indicates that excessive perturbation may disturb the stability of the learned features.Table 4.2 provides the class-wise evaluation of the classification results.

Chapter 6

CONCLUSIONS AND SCOPE FOR FURTHER WORK

In this work, a chaos-enhanced few-shot learning framework was developed for brain tumor classification based on medical image data. This model uses a combination of a convolutional network and a prototypical network to classify images. Instead of utilizing conventional data augmentation methods, a feature perturbation based on chaos was employed to improve the robustness of the model. The experimental results reveal that incorporating a certain level of chaos in the feature embedding space improves the performance of the classification model. This is because a moderate level of chaos helps the model learn more discriminative features. This improves the classification accuracy. However, the results reveal that a moderate level of chaos is required to achieve optimal performance. Excessive levels of chaos have a detrimental effect on classification accuracy. This is due to the unstable nature of the feature distribution.Overall, the proposed method demonstrates the efficacy of using a combination of chaos theory and few-shot learning in medical image classification problems, where the amount of training data is limited. The framework offers a promising direction for enhancing deep learning models without relying on large datasets or conventional augmentation techniques.

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